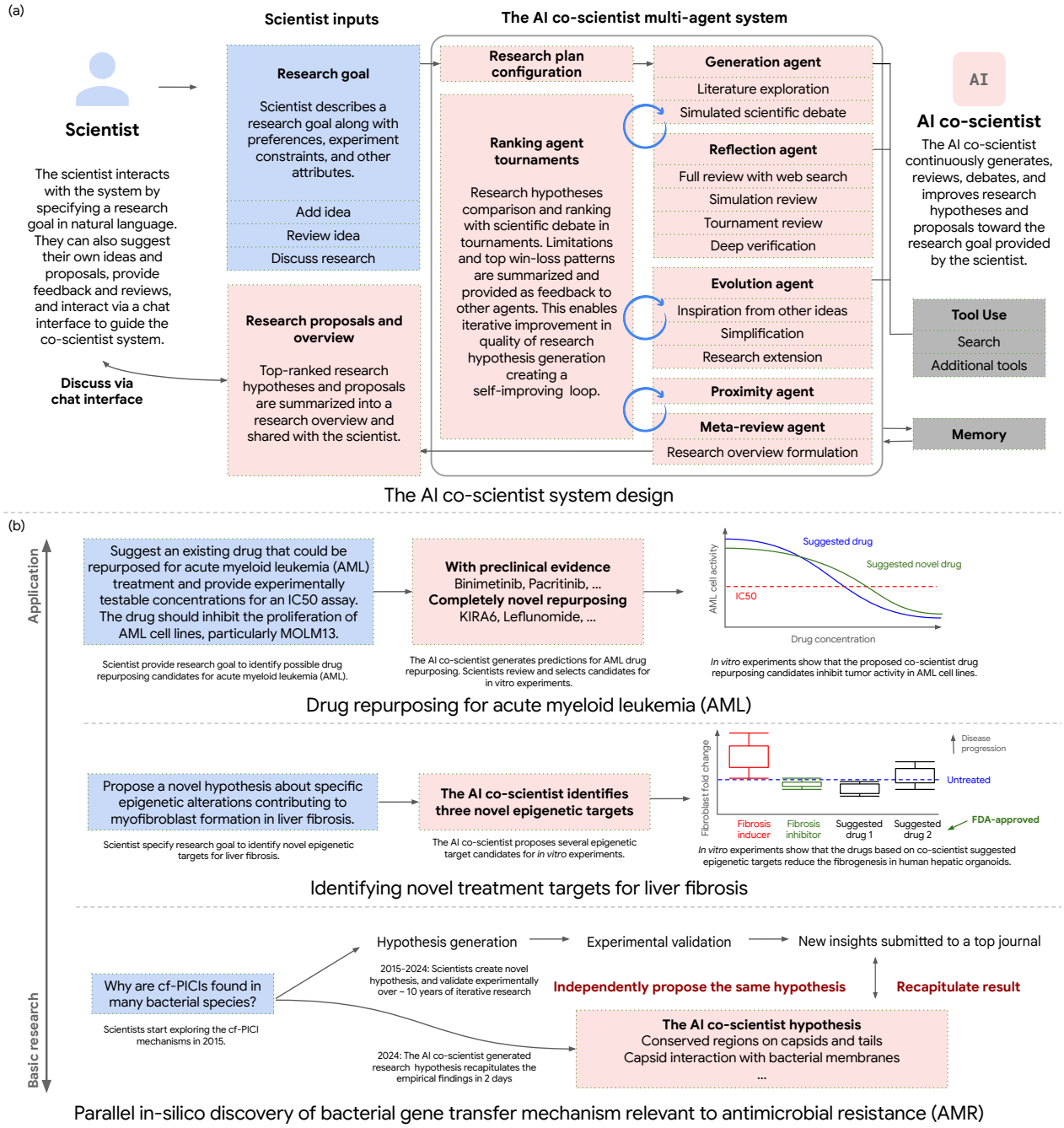

Figure 1 | The AI co-scientist system design and experimental validation summary. (a) Here, we illustrate the different components of the the AI co-scientist multi-agent system, and its interaction paradigm with scientists. Given a research goal in natural language, the co-scientist generates novel research hypotheses and proposals. The system employs specialized agents — Generation, Reflection, Ranking, Evolution, Proximity (which evaluates relatedness), Meta-review (which provides high level analysis) — to continuously generate, debate, and evolve research hypotheses within a tournament framework. Feedback from the tournament enables iterative improvement, creating a self-improving loop towards novel and high-quality outputs. The co-scientist leverages tools, including web search and specialized AI models to improve the grounding and quality of generated research hypotheses. Scientists can converse with the co-scientist in natural language to specify research goals, incorporate constraints, provide feedback and suggest new directions for explorations via the designated user interface. (b) We perform end-to-end validation of the co-scientist generated hypotheses in three important topics of biomedicine with varied complexity—suggesting novel drug repurposing candidates for acute myeloid leukemia (AML) (upper panel), discovering novel epigenetic targets for liver fibrosis treatment (middle panel), and recapitulating the discovery of novel mechanism of gene transfer evolution in bacteria key to anti-microbial resistance (lower panel). The co-scientist's hypotheses for these three settings are externally, independently validated by *in vitro* laboratory experiments and detailed in separate preprints co-timed with this work. In the figure, blue denotes expert scientist inputs while red denotes the co-scientist agents or outputs.

There has been rapid technological progress in AI towards generally intelligent and collaborative systems, which might empower scientists in creatively traversing and expertly reasoning across disciplinary domains. Such systems are capable of advanced reasoning [4–6], multimodal understanding [6], and agentic behaviors [7] such as the ability to use tools to solve complex tasks over long time horizons. Further, the trends with distillation [8] and inference time compute costs [6, 9] indicate that such intelligent and general AI systems are rapidly becoming more affordable and available. Motivated by the aforementioned unmet needs in the modern discovery process in science and medicine and building on the advancements in frontier AI [10], we develop and introduce an AI co-scientist system.

The co-scientist is designed to act as a helpful assistant and collaborator to scientists and to help accelerate the scientific discovery process. The system is a compound, multi-agent AI system [11] building on Gemini 2.0 and designed to mirror the reasoning process underpinning the scientific method [12]. Given a research goal specified in natural language, the system can search and reason over relevant literature to summarize and synthesize prior work and build on it to propose novel, original research hypotheses and experimental protocols for downstream validations (Figure 1a). The co-scientist provides grounding for its recommendations by citing relevant literature and explaining the reasoning behind its proposals.

This work does not aim to completely automate the scientific process with AI. Instead, the co-scientist is purpose-built for a "scientist-in-the-loop" collaborative paradigm, to help domain experts augment their hypothesis generation process and guide the exploration that follows. Scientists can specify their research goals in simple natural language, including informing the system of desirable attributes for the hypotheses or research proposals it should create and the constraints that the synthesized outputs should satisfy. They can also collaborate and provide feedback in a variety of ways, including directly supplying their own ideas and hypotheses, refining those generated by the system, or using natural language chat to guide the system and ensure alignment with their expertise.

The co-scientist works through a significant scaling of the test-time compute paradigm [13–15] to iteratively reason, evolve, and improve the outputs as it gathers more knowledge and understanding. Underpinning the system are thinking and reasoning steps—notably a self-play based scientific debate step for generating novel research hypotheses; tournaments that compare and rank hypotheses via the process of finding win and loss patterns, and a hypothesis evolution process to improve their quality. Finally, the agentic nature of the system enables it to recursively self-critique its output and use tools such as web-search to provide itself with feedback to iteratively refine its hypotheses and research proposals.

While the co-scientist system is general-purpose and applicable across multiple scientific disciplines, in this study we focus our development and validation of the system to biomedicine. We validate the co-scientist's capability in three impactful areas of biomedicine with varied complexity: (1) drug repurposing, (2) novel treatment targets discovery, and (3) new mechanistic explanations for antimicrobial resistance (Figure 1b).

Drug development is an increasingly time-consuming and expensive process [16] in which new therapeutics require restarting many aspects of the discovery and development process for each indication or disease (roughly 70% of drug approvals are for new drugs). In contrast, drug repurposing——identifying novel therapeutic indications for drugs beyond their original intended use——has emerged as a compelling strategy to address these challenges [17]. Successful examples of repurposing include Humira (adalimumab) and Keytruda (pembrolizumab), both of which have become among the most successful drugs in history. [17]. The process typically involves analyzing molecular signatures, signaling pathways, drug interactions, clinical trial results, adverse event reports, and other literature-based information [18], along with off-label use data and, in some cases, patient experiences. However, drug repurposing is limited by several factors: (1) the need for extensive expertise across biomedical, molecular biology, and biochemical systems, (2) the inherent complexity of mammalian biological systems, and (3) the time-intensive nature of traditional computational biology analyses required. We leverage the co-scientist to generate predictions for large-scale drug repurposing, validating the generated predictions using a combination of computational biology, expert clinician feedback, and *in vitro* wet-lab validation approaches. Notably, our system has proposed novel repurposing candidates for acute myeloid leukamia (AML) that inhibit tumor viability at clinically relevant concentrations *in vitro* across multiple AML cell lines.

Unlike drug repurposing, which is a combinatorial search problem through a large but constrained set of

drugs and diseases, identifying novel treatment targets for diseases presents a more significant challenge, traditionally requiring extensive literature review, deep biological understanding, sophisticated hypothesis generation and complex experimental validation strategies. The uncertainty of identifying novel treatment targets is significantly greater than in drug repurposing, as it involves not only repurposing existing compounds but also uncovering entirely new components and mechanisms within biological systems. This target discovery process can be inefficient, potentially leading to suboptimal hypothesis selection and prioritization for *in vitro* and *in vivo* experimentation. Given the high costs and time associated with experimental validation, a more effective approach is needed. We probe the capabilities of the co-scientist to propose, rank, and provide experimental protocols for novel research hypotheses pertaining to target discovery. To demonstrate this capability, we focus on liver fibrosis, a prevalent and serious disease, showcasing the co-scientist's potential to discover novel treatment targets amenable to experimental validation. In particular, the co-scientist has suggested novel epigenetic targets demonstrating significant anti-fibrotic activity in human hepatic organoids.

As a third validation of the capabilities of our system, we focus on generation of hypotheses to explain mechanisms related to gene transfer evolution in bacteria pertaining to antimicrobial resistance (AMR) - mechanisms developed by microbes to circumvent drug applications used to fight infections. This is arguably an even more complex challenge than drug repurposing and target discovery and involves understanding of not only the molecular mechanisms of gene transfer (conjugation, transduction, and transformation) but also the ecological and evolutionary pressures that drive the spread of AMR genes: a system-level problem with many interacting variables. This is also an important healthcare challenge with increasing rates of infections and deaths worldwide [19]. In this validation, researchers instructed the AI co-scientist to explore a topic that had already been subject to novel discovery by their independent research group. Notably, at the time of instructing the AI co-scientist system, the researchers' novel experimental insights had not yet been published or revealed in the public domain. The system was instructed to hypothesize how capsid-forming phage-inducible chromosomal islands (cf-PICIs) exist across multiple bacterial species. The system independently proposed that cf-PICIs interact with diverse phage tails to expand their host range. This *in silico* discovery mirrored the novel and experimentally validated results that expert researchers had already performed, as detailed in the co-timed report [20, 21].

Overall, our key contributions are summarized as follows:

- **Introducing an AI co-scientist.** We develop and introduce an AI co-scientist that goes beyond literature summarization and "deep research" tools to assist scientists in uncovering new knowledge, novel hypothesis generation and experimental planning.
- **Significant scaling of test-time compute paradigm for scientific reasoning.** The co-scientist is built on a Gemini 2.0 multi-agent architecture, utilizing an asynchronous task execution framework. This framework allows the system to flexibly allocate computational resources to scientific reasoning, mirroring key aspects of the scientific method. Specifically, the system uses self-play strategies, including a scientific debate and a tournament-based evolution process, to iteratively refine hypotheses and research proposals creating a self-improving loop. Using automated evaluations across 15 complex expert curated open scientific goals, we demonstrate the benefits of scaling the test-time compute paradigm with the AI co-scientist outperforming other state-of-the-art (SOTA) agentic and reasoning models in generating high quality hypotheses for complex problems.
- **Expert-in-the-loop scientific workflow.** Our system is designed for collaboration with scientists. The system can flexibly incorporate conversational feedback in natural language from scientists and co-develop, evolve and refine outputs.
- **End-to-end validation of the co-scientist in important topics in biomedicine.** We present end-to-end validation of novel AI-generated hypotheses through new empirical findings in three distinct and increasingly complex areas of biomedicine: drug repurposing, novel target discovery, and antimicrobial resistance. The AI co-scientist predicts novel repurposing drugs for AML, identifies novel epigenetic treatment targets grounded in preclinical evidence for liver fibrosis, and proposes novel mechanisms for gene transfer in bacterial evolution and antimicrobial resistance. These discoveries from the AI co-scientist have been validated in wet-lab settings and are detailed in separate, co-timed technical reports.

2 Related Works

2.1 Reasoning models and test-time compute scaling

The modern revolution in foundation AI models [22] and large language models (LLMs) has been largely driven by advances in pre-training techniques [23, 24], leading to breakthroughs in models like the GPT and Gemini family [25, 26]. These models, trained on increasingly massive internet-scale and multimodal datasets, have demonstrated impressive abilities in language understanding and generation leading to breakthrough performance in a variety of benchmarks [27, 28]. However, a key area of ongoing development is enhancing their *reasoning* capabilities. This has led to the emergence of "reasoning models" which go beyond simply predicting the next word and instead attempt to mimic human thought processes [29]. One promising direction in this pursuit is the test-time compute paradigm. This approach moves beyond solely relying on the knowledge acquired during pre-training and allocates additional computational resources during inference to enable System-2 style thinking—slower deliberate reasoning to reduce uncertainty and progress optimally towards the goal [30]. This concept emerged with early successes such as AlphaGo [15], which used Monte Carlo Tree Search (MCTS) to explore game states and strategically select moves, and Libratus [14], which employed similar techniques to achieve superhuman performance in poker. This paradigm has now found applications in LLMs, where increased compute at test-time allows for more thorough exploration of possible responses, leading to improved reasoning and accuracy [11, 13, 29, 31–35]. Recent advancements, like the Deepseek-R1 model [4], further demonstrate the potential of test-time compute by leveraging reinforcement learning to refine the model's "chain-of-thought" and enhance complex reasoning abilities over longer horizons. In this work, we propose a significant scaling of the test-time compute paradigm using inductive biases derived from the scientific method to design a multi-agent framework for scientific reasoning and hypothesis generation without any additional learning techniques.

2.2 AI-driven scientific discovery

AI-driven scientific discovery represents a paradigm shift in how research is conducted across various scientific domains. Recent advancements, particularly the development of large deep learning and generative models, have cemented AI's role in scientific discovery. This is best exemplified by AlphaFold 2's remarkable progress in the grand challenge of protein structure prediction, which has revolutionized structural biology and opened new avenues for drug discovery and materials science [36]. Other notable examples include the development of novel antibiotics, protein binder design, and material discovery with AI [37–39].

Building on these successes with specialized, bespoke AI models, there has been recent work exploring the even more ambitious goal of fully integrating AI, especially modern LLM-based systems, into the complete research workflow, from initial hypothesis generation all the way to manuscript writing. This end-to-end integration represents a significant shift, presenting both unprecedented opportunities and significant challenges as the field moves beyond specialized AI tools toward realizing the potential of AI as an active collaborator, or even, as some envision, a nascent "AI scientist" [40, 41].

As an example of this shift, Liang et al. [42] directly assessed the utility of LLMs for providing feedback on research manuscripts. Through both a retrospective analysis of existing peer reviews and a prospective user study, they demonstrated the significant concordance between LLM-generated feedback and that of human reviewers. Their study, using GPT-4 [43], found that a majority of researchers perceived LLM-generated feedback as helpful, and in some instances, even more beneficial than feedback from human colleagues. However, while valuable, their work focuses solely on the feedback stage of the scientific process, leaving open the question of how LLMs might be integrated into the full research cycle, from hypothesis formation to experimental validation and manuscript writing.

Another effort embodying this shift is PaperQA2 [44], an AI agent for scientific literature search and summarization. The authors claimed to surpass PhD and postdoc researchers on multiple literature research tasks, as measured both by performance on objective benchmarks and human evaluations. While the system is a useful for synthesizing information, it does not engage in scientific reasoning for novel hypothesis generation.

HypoGeniC, a system proposed by Zhou et al. [45], tackles hypothesis generation by iteratively refining hypotheses using LLMs and a multi-armed bandit-inspired approach. The process begins with a small set of

examples, from which initial hypotheses are generated. These hypotheses are then iteratively updated through exploration and exploitation, guided by a reward function based on training accuracy. This refined set of hypotheses is subsequently used to construct an interpretable classifier. However, the method's reliance on retrospective data for evaluation means the degree to which the system can generate truly novel hypotheses remains an open question. Furthermore, the system lacks end-to-end validation beyond subjective human evaluations.

Ifargan et al. [46] present "data-to-paper", a platform that systematically guides multiple LLM and rule-based agents to generate research papers, with automated feedback mechanisms and information tracing for verification. However, the evaluations are limited to recapitulating existing peer-reviewed publications and its unclear if the system can generate truly novel, yet grounded hypothesis and research proposals.

Virtual Lab [47] is another closely related work. Here, the authors propose a team of LLM agents with a "principal investigator" LLM guiding a team of specialized LLM agents to solve a scientific problem. The LLM team receives high level human supervision. The authors demonstrate the utility of their work by leveraging Virtual Lab to design nanobody binders to recent variants of SARS-CoV-2 with experimental validation. While similar in spirit, there are significant design differences to our approach and the generality of the system remains unclear.

Boiko et al. [48] introduced "Coscientist", a multi-agent system powered by GPT-4, designed for autonomous execution of complex chemical experiments. This system integrates capabilities such as web and document searching, and code execution, to facilitate independent experimental design, planning, and execution. In addition to similar sounding names, both "Coscientist" and our system share the overarching goal of accelerating scientific discovery through AI. However, there are several important distinctions. Notably, "Coscientist" is quite narrowly focused on chemical research while ours is much broadly applicable across science. Secondly, our system has important technical innovations that lead to a self-improving system that can uncover new, original knowledge while their approach is a more vanilla-stitching of GPT-4 based agents. Finally, despite the name, "Coscientist" prioritizes a high degree of autonomy in experimental execution, directly interfacing with laboratory hardware. Our system, instead, is explicitly designed as a collaborative tool, emphasizing a "scientist-in-the-loop" approach and centers on the more cognitive aspects of the research process.

Finally, Lu et al. [40] propose "The AI Scientist", a fully automated system designed to conduct research using multiple collaborating LLM agents. These agents handle all stages of the research process, from defining research problems and conducting literature reviews to designing and executing experiments, and even writing up the results. The design shares similarities with our work—the key differences being our focus on the scaling of the test-time compute paradigm to generate high quality hypotheses and research proposals. Secondly, their proposed system has limited automated evaluations; in contrast, our work has a combination of automated, human expert and end-to-end wet lab validations. Finally, our goal is to not to automate scientific discovery, rather to build a helpful AI collaborator for scientists.

2.3 AI for biomedicine

More broadly, large AI models are increasingly demonstrating their potential in biomedical science. Both general purpose (GPT-4, Gemini) and specialized LLMs (Med-PaLM, Med-Gemini, Galactica, Tx-LLM) have shown strong performance on biomedical reasoning and question-answering benchmarks [25, 26, 49–52]. Beyond benchmarks, Med-PaLM 2, was successfully applied to identify causative murine genetic factors for traits such as diabetes, cataracts, and hearing loss [53]—an early example of hypothesis generation and LLM-assisted discovery. We have also seen the exciting development of specialized foundation and large language models trained on DNA, RNA and protein sequences with a variety of applications [54–57]. Although AI in biology and medicine often necessitates specialization, the rapid progress of frontier AI models has blurred the distinction. As these models grow in scale, data diversity, and complexity, they continue to achieve breakthroughs in areas once thought to require domain-specific AI. Our co-scientist system, with its modular multi-agent architecture, is flexibly designed to build on top of these advancements in general-purpose frontier AI models and leverage specialized AI models as tools to enhance the capabilities.

Drug repurposing is an important area of validation experiments in this work. The traditional approach to this task requires both computational and experimental approaches and a comprehensive understanding of

disease-drug interactions [17, 58]. While methods like knowledge graphs with graph convolutional networks have shown promise [59, 60], their applicability is limited by the initial knowledge graph's scope. TxGNN [61], an example of a specialized biomedical foundation model with a graph based approach, addresses "zero-shot" repurposing for novel diseases but remains dependent on the underlying knowledge graph's quality and lacks sufficient scalability and explainability. Furthermore, no end-to-end validations of the model predictions were reported in the study. In contrast, our work, leveraging state-of-the-art LLMs in the co-scientist setup, is more scalable. We report a combination of expert evaluations and wet-lab experiments to validate the system predictions.

3 Introducing the AI co-scientist

This section describes the technical details, agents, and framework comprising the co-scientist system. The co-scientist employs a multi-agent architecture built upon Gemini 2.0, integrated within an asynchronous task execution framework. This framework allows for flexible scaling of test-time compute resources, facilitating advanced scientific reasoning.

Given a research goal specified by an expert scientist in natural language, the co-scientist generates hypotheses and research proposals that adhere to the following default criteria:

- **Alignment with the provided research goal.** The generated outputs must precisely align with the research goals, preferences and constraints defined by the scientist.
- **Plausibility.** The system outputs should be free of readily apparent flaws. Any potential contradictions with prior literature or established knowledge must be explicitly stated and justified.
- **Novelty.** A key objective of the co-scientist system is to generate novel hypotheses, conjectures, and research plans grounded in prior literature, rather than simply synthesizing existing information (a capability already addressed by existing "deep research" tools [62]).
- **Testability.** The system outputs should be amenable to empirical validation within the constraints specified by the scientist.
- **Safety.** The system outputs will be controlled to prevent enabling unsafe, unethical, or harmful research.

Aside from these default criteria, the co-scientist can be configured with additional criteria, preferences, and constraints as needed. For instance, it can be configured to generate outputs in formats preferred by the researcher to improve interpretability and readability.

Throughout this section, we employ a recurring example: generating hypotheses for exploring the biological mechanisms of Amyotrophic Lateral Sclerosis (ALS) to illustrate the various components of the co-scientist system. While this example has been reviewed by domain experts, it remains illustrative and may contain errors. Importantly, this example does not aim to suggest potential therapeutic avenues for ALS and should be interpreted with utmost caution. All the examples are listed in the Appendix Section A.3.

3.1 The AI co-scientist system overview

At a high level, the co-scientist system comprises four key components:

- **Natural language interface.** Scientists interact with and supervise the system primarily through natural language. This allows them to not only define the initial research goal but also refine it at any time, provide feedback on generated hypotheses (including their own solutions), and generally guide the system's progress.
- **Asynchronous task framework.** The co-scientist employs a multi-agent system where specialized agents operate as worker processes within an asynchronous, continuous, and configurable task execution framework. A dedicated Supervisor agent manages the worker task queue, assigns specialized agents to these processes, and allocates resources. This design enables the system to flexibly and effectively utilize computational resources and iteratively improve its scientific reasoning capabilities.
- **Specialized agents.** Following inductive biases and scientific priors derived from the scientific method, the process of scientific reasoning and hypothesis generation is broken down into sub-tasks. Individual,

specialized agents, each equipped with customized instruction prompts, are designed to execute these sub-tasks. These agents operate as workers coordinated by the Supervisor agent.

- **Context memory.** In order to enable iterative computation and scientific reasoning over long time horizons, the co-scientist uses a persistent context memory to store and retrieve states of the agents and the system during the course of the computation.

The Gemini 2.0 model is the foundational LLM underpinning all agents in the co-scientist system. The specific co-scientist design was arrived at with iterative developments and is reflective of the current capabilities of the underlying LLMs.

3.2 From research goal to research plan configuration

The research goal, specified by the scientist, serves as the entry point to the co-scientist system. Leveraging the multimodal and long context capabilities of Gemini 2.0 models, the co-scientist efficiently processes research goals of varying complexity, from simple statements to extensive documents spanning tens of thousands of natural language tokens or other relevant data (e.g., including hundreds of prior publication PDFs). The research goal may also incorporate specific constraints, attributes, and preferences related to the scientist's particular laboratory setting or field of work.

The co-scientist system then parses the goal to derive a research plan configuration for generating research proposals. This configuration captures the desired proposal preferences, attributes, and constraints. For example, it specifies whether the co-scientist should exclusively propose novel hypotheses. It also specifies the criteria for evaluating hypothesis quality, such as novelty and experimental feasibility. These criteria are then used by the system during its auto-evaluation and improvement phases. The attributes, preferences, and evaluation criteria can all be customized to a given research goal. To illustrate this process, we present an example research goal and its corresponding parsed research plan configuration in Appendix Figure A.9, where the goal is to develop a novel hypothesis related to phosphorylation of the Nuclear Pore Complex (NPC) as a causative mechanism for ALS [63].

Based on the research plan configuration, the Supervisor agent initiates the creation of a task queue and begins orchestrating the specialized agents. The system operates continuously and asynchronously. Periodically, the Supervisor agent calculates a comprehensive set of summary statistics, reflecting the system's state and progress toward the specified research goal. These statistics inform decisions regarding resource allocation and the determination of whether a terminal state for the overall computation has been reached. The state is periodically written to the associated context memory of the system and leveraged as feedback in subsequent rounds of computation. It also enables easy restarts in-case of any failure in the system components.

3.3 The specialized agents underpinning the AI co-scientist

At the core of the co-scientist system are a coalition of specialized agents, each orchestrated by the Supervisor agent. These agents are designed to emulate the scientific reasoning process, enabling them to generate novel hypotheses and research plans. They are also equipped to interact with external tools, such as web search engines and specialized AI models, through application programming interfaces (APIs). These specialized agents are enumerated below:

- **Generation agent.** The agent initiates the research process by generating the initial focus areas, iteratively extending them and generating a set of initial hypotheses and proposals that address the research goal. This involves exploring relevant literature using web search, synthesizing existing findings into novel directions, and engaging in simulated scientific debates for iterative improvement.
- **Reflection agent.** This agent simulates the role of a scientific peer reviewer, critically examining the correctness, quality, and novelty of the generated hypotheses and research proposals. Furthermore, it evaluates the potential of each hypothesis to provide an improved explanation for existing research observations (identified via literature search and review), particularly those that may be under explained.
- **Ranking agent.** An important abstraction in the co-scientist system is the notion of a tournament where different research proposals are evaluated and ranked enabling iterative improvements. The Ranking agent employs and orchestrates an Elo-based tournament [64] to assess and prioritize the

generated hypotheses at any given time. This involves pairwise comparisons, facilitated by simulated scientific debates, which allow for a nuanced evaluation of the relative merits of each proposal.

- **Proximity agent.** This agent asynchronously computes a proximity graph for generated hypotheses, enabling clustering of similar ideas, de-duplication, and efficient exploration of the hypothesis landscape.
- **Evolution agent.** The co-scientist's iterative improvement capability relies heavily on this agent, which continuously refines the top-ranked hypotheses emerging from the tournament. Its refinement strategies include synthesizing existing ideas, using analogies, leveraging literature for supporting details, exploring unconventional reasoning, and simplifying concepts for clarity.
- **Meta-review agent.** This agent also enables the co-scientist's continuous improvement by synthesizing insights from all reviews, identifying recurring patterns in tournament debates, and using these findings to optimize other agents' performance in subsequent iterations. This also enhances the quality and relevance of generated hypotheses and reviews in subsequent iterations. The agent also synthesizes top-ranked hypotheses and reviews into a comprehensive research overview for review by the scientist.

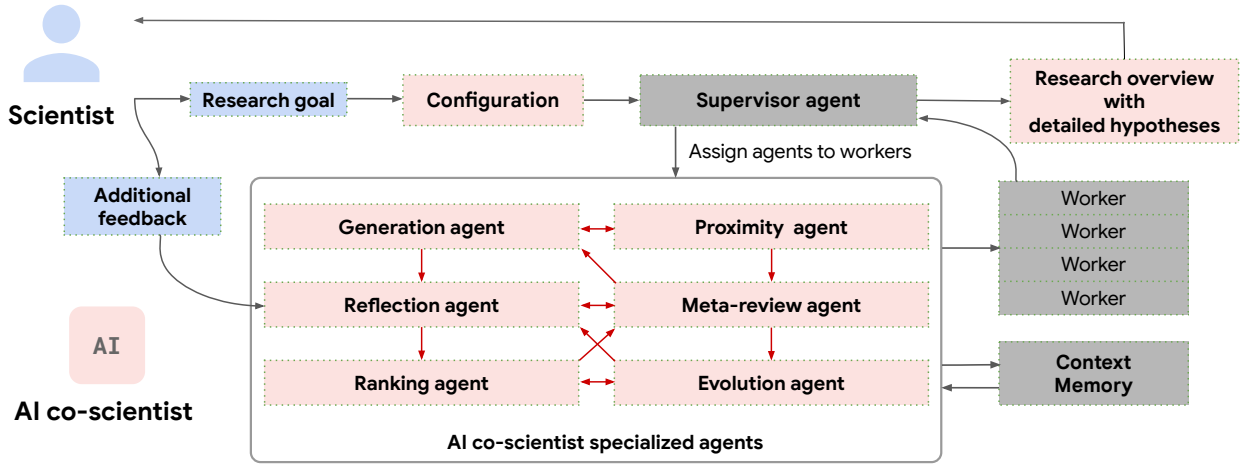

Figure 2 | The AI co-scientist multi-agent architecture design. The co-scientist accepts a natural language research goal from the user and parses this into a research plan configuration. This plan is then dispatched to the Supervisor agent which evaluates this plan to assigns weights and resources to each specialized agent and subsequently queues them as worker processes in a task queue according to these weights. The worker processes execute the queue of agent actions, and the system ultimately aggregates all information to formulate a research overview with detailed hypotheses and proposals for the scientist. The red boxes in the "The AI co-scientist specialized agents" section denote individual agents each with their own unique logic and role. The blue boxes indicate the scientist-in-the-loop inputs and feedback. The dark gray arrows represent the information flow through the co-scientist system, while the red arrows represent the information feedback loop between the specialized agents.

The Supervisor agent's seamless orchestration of these specialized agents enables the development of valid, novel, and testable hypotheses and research plans tailored to the input research goal.

In summary, the Generation agent curates an initial list of research hypotheses satisfying a research goal. These are then reviewed by the Reflection agent and evaluated in a tournament by the Ranking agent. The Evolution, Proximity, and Meta-review agents operate on the tournament state to help improve the quality of the system outputs.

The Supervisor agent periodically computes and writes to the context memory, a comprehensive suite of statistics, including the number of hypotheses generated and requiring review, and the progress of the tournament. These statistics also include analyses of the effectiveness of different hypothesis generation methodologies (e.g., generating new ideas via the Generation agent vs. improving existing ideas via the Evolution agent). Based on these statistics, the Supervisor agent then orchestrates subsequent system operations, i.e., generating new hypotheses, reviews, tournaments, and improvements to existing hypotheses, by strategically weighting and sampling the specialized agents for execution via the worker processes.

Importantly, the Meta-review agent enables feedback propagation and learning without back-propagation techniques (e.g., fine-tuning or reinforcement learning) [65]. The Meta-review agent generates feedback

applicable to all agents, which is simply appended to their prompts in the next iteration—a capability facilitated by the long-context search and reasoning capabilities of the underlying Gemini 2.0 models. Through this feedback loop, the co-scientist continuously learns and improves in subsequent iterations with more compute scaling.

Finally, while our work leverages Gemini 2.0, the co-scientist framework is model-agnostic and portable to other similar models or combinations thereof. Future LLM improvements will likely enhance the co-scientist's capabilities. The multi-agent architecture of the co-scientist is depicted and summarized in Figure 2.

We now describe the mechanisms of action of the specialized agents in more detail.

3.3.1 Generation agent

The co-scientist Generation agent employs a diverse array of techniques and tools to generate novel hypotheses, such as the following:

- **Literature exploration via web search.** The agent iteratively searches the web, retrieves and reads relevant research articles, and grounds its reasoning by summarizing prior work. It then builds on this summary to generate novel hypotheses and research plans. An example prompt is given in Appendix Figure A.1.
- **Simulated scientific debates.** Here, the Generation agent simulates scientific debates among experts by employing self-critique and self-play techniques. These debates typically involve multiple turns of conversations leading to a refined hypothesis generated at the end. An example prompt is given in Appendix Figure A.2.
- **Iterative assumptions identification.** The agent iteratively identifies testable intermediate assumptions, which, if proven true, can lead to novel scientific discovery. These plausible assumptions and their sub-assumptions are identified through conditional reasoning hops and subsequently aggregated into complete hypotheses.
- **Research expansion.** To identify previously unexplored areas of the hypothesis space, the Generation agent reviews existing hypotheses and the research overview and feedback provided by the Meta-review agent in the previous iteration. This is used to inform additional exploration directions in the research hypothesis space.

An example hypothesis and research proposal output from the Generation agent is presented in Appendix Figure A.10 for the aforementioned research goal regarding explaining a basic mechanism related to ALS. The Generation agent also summarizes and categorizes each generated hypothesis, allowing scientists to quickly grasp the core ideas.

3.3.2 Reflection agent

Reviews are integral to the co-scientist's effectiveness in generating novel proposals. The Reflection agent searches relevant prior work (via web search or a dedicated scientist-provided repository), assesses existing experimental evidence for or against a given hypothesis, and rigorously verifies the novelty, correctness, and quality of generated outputs. Effective reviews filter inaccurate and, when stipulated, non-novel hypotheses. Moreover, they also provide feedback to all other agents, driving continuous improvement. The Reflection agent employs the following types of review:

- **Initial review.** Building on the co-scientist's default evaluation criteria, the Reflection agent performs an initial review assessing the correctness, quality, novelty, and a preliminary assessment of safety (ethics) of the generated hypotheses. For a more in-depth discussion on safety considerations see Section 6. This initial review, which doesn't use external tools like web search, aims to quickly discard flawed, non-novel, or otherwise unsuitable hypotheses.
- **Full review.** If a hypothesis passes the initial review, the Reflection agent performs a full review, leveraging external tools and web searches to identify relevant articles for improved reasoning and grounding. This review evaluates the hypothesis's correctness, quality, and novelty similar to the initial review but with full literature search. For correctness and quality, the agent scrutinizes underlying

assumptions and reasoning. For novelty, it summarizes known aspects of the hypothesis and then judges their novelty based on existing literature. An example full novelty review is shown in Appendix Figure A.11, and an example of review critiques is in Appendix Figure A.12. A complete full review example is shown in Appendix Figure A.13.

- **Deep verification review.** The Reflection agent also conducts a deep verification review, decomposing the hypothesis into constituent assumptions. Each assumption is further broken down into fundamental sub-assumptions, decontextualized, and independently evaluated for correctness to identify invalidating elements for subsequent filtering. Concurrently, the reasons for potential hypothesis invalidation due to incorrect assumptions are summarized. This deep verification helps the co-scientist detect subtle errors within complex hypotheses, such as flaws in reasoning or inaccurate experimental protocols. An identified error doesn't necessarily invalidate the core hypothesis; the Reflection agent assesses whether the incorrect assumption is fundamental to the hypothesis and incorporates this reasoning into the review. Non-fundamental errors can be addressed during subsequent refinement stages. An example deep verification review is provided in Appendix Figure A.14 for the previously introduced ALS hypothesis. We also show another example of a deep verification review via probing questions in the context of drug repurposing for AML in Appendix Figure A.15.
- **Observation review.** In addition, the Reflection agent also explores whether a given hypothesis can account for long-tail observations from prior experimental results. This review aims to determine if the hypothesis can provide insights on existing experimental findings and observed phenomena within relevant articles. For each observation, the agent assesses if the hypothesis is a superior explanation over existing ones, assuming its validity. Positive observations are summarized and appended to the hypothesis. Note that this review often completes without any important findings (as in the case of the ALS hypothesis example). An example prompt to generate observations is provided in Appendix Figure A.3. An illustrative example of an observation review is provided in Appendix Figure A.16 in the context of an alternate hypothesis for explaining a mechanism of anti-microbial resistance.
- **Simulation review.** The Reflection agent also reviews hypotheses by simulating them in a step-wise fashion (e.g., simulating the mechanism of action or the proposed experiment in the proposal). This simulation allows the agent to identify and summarize potential failure scenarios. This review method is built on the assumption that frontier LLMs may have developed an internal world model that enables them to simulate and accurately predict various scientific phenomena.
- **Recurrent/tournament review.** The Reflection agent adapts its full reviews based on the co-scientist's growing knowledge. By analyzing reviewed hypotheses and results of the tournament conducted by the Ranking agent, the Reflection agent identifies recurring issues and improvement opportunities, refining its reviews accordingly.

Additionally, the co-scientist can incorporate reviews from expert scientists to guide ranking and improvements (further discussed in Section 3.4). We aim to have the Reflection agent's comprehensive set of reviews cover the common methods scientists employ when critiquing and refining research hypotheses and proposals.

3.3.3 Ranking agent

The AI co-scientist explores numerous hypotheses and research proposals towards a research goal, necessitating a ranking mechanism to prioritize computational resources toward the most promising candidates. This task is performed by the Ranking agent. The agent uses an Elo-based tournament [64] to automatically evaluate and rank all hypotheses, providing supporting rationale. This ranking serves to communicate to scientists an ordered list of research hypotheses and proposals aligned with the research goal. Despite its assumptions and limitations [66], Elo remains a good proxy for relative ranking, and it has previously been applied to rank extracted patterns and ideas in games [67]. In the future, extensions may be considered [68]. We set the initial Elo rating of 1200 for the newly added hypothesis.

Because the tournament is computationally intensive, the Ranking agent employs several optimization strategies. Top-ranked hypotheses are compared pairwise in tournament matches through multi-turn scientific debates [69]. This mitigates ordering bias and focuses on novelty, correctness, and testability. Lower-ranked hypotheses undergo single-turn comparisons in a pairwise fashion in their tournament match. The agent concludes each comparison with a decision regarding which hypothesis is better. Appendix Figure A.4 and

Appendix Figure A.5 show example prompts. Appendix Figure A.17 shows an example of the Ranking agent conducting a scientific debate match in a tournament to compare two hypotheses.

The Ranking agent prioritizes tournament matches as follows: (1) hypotheses are more likely to be compared with similar ones (based on the Proximity agent's graph, described in the next section); (2) newer and top-ranking hypotheses are prioritized for participation in tournament matches. Successful hypotheses quickly achieve favorable rankings and this informs the tournament state for subsequent iterations.

3.3.4 Proximity agent

The Proximity agent calculates the similarity between research hypotheses and proposals, and builds a proximity graph, taking into account the specific research goal. Although it doesn't directly participate in hypothesis generation, the Proximity agent assists the Ranking agent in organizing tournament matches and showcasing a diverse range of ideas related to the research goal. This allows scientists to quickly explore areas of interest and easily identify related concepts.

3.3.5 Evolution agent

The Evolution agent continuously refines and improves existing hypotheses and proposals using several approaches including:

- **Enhancement through grounding.** Here the agent attempts to improve hypotheses by identifying weaknesses, generating search queries, retrieving and reading articles, suggesting improvements and elaborating on details to fill reasoning gaps.
- **Coherence, practicality and feasibility improvements.** The agent aims to address issues and creates more coherent hypotheses, potentially rectifying underlying problems with invalid initial assumptions. The agent also refines the hypotheses to make them more practical and feasible. Appendix Figure A.6 provides an example of the feasibility improvement prompt.
- **Inspiration from existing hypotheses.** The agent additionally creates new hypotheses inspired by single or multiple top-ranked hypotheses.
- **Combination.** The agent also attempts to directly combine the best aspects of several top-ranking hypotheses to create new hypotheses.
- **Simplification.** The agent simplifies hypotheses for easier verification and testing.
- **Out-of-box thinking.** The agent also explores out-of-the-box ideas by moving away from a subset of hypotheses and generating divergent ones. Appendix Figure A.7 provides an example prompt for this.

The Evolution agent generates new hypotheses; it doesn't modify or replace existing ones. This strategy protects the quality of top-ranked hypotheses from flawed improvements, as each new hypothesis must also compete in the tournament. The evolution of research hypotheses and proposals also allows the co-scientist to iteratively combine different improvement techniques and gradually improve the quality of the results.

3.3.6 Meta-review agent

The Meta-review agent plays a crucial role in the co-scientist's feedback loop, enabling self-improvement in scientific reasoning. This agent operates on the tournament state and summarizes common patterns identified in reviews and scientific debates in the tournament matches into a meta-review critique.

By synthesizing insights from all reviews, the meta-review provides valuable feedback to the Reflection agent, leading to more thorough and reliable future reviews. This helps prevent oversight of critical details. Consider the illustrative example of a identifying a repurposing drug candidate for ALS as a research goal: while only 90% of individual reviews might correctly identify a blood-brain barrier permeability issue in a proposed candidate, the meta-review ensures that all future reviews by the Reflection Agent definitively address this crucial factor. Hypothesis and research proposal generation is also enhanced by the meta-review's identification of recurring issues. While the Generation agent uses this feedback selectively to avoid over fitting to these review critiques, it helps prevent the recurrence of common issues.

Appendix Figure A.8 provides an example prompt for the meta-review. In Appendix Figure A.18-A.19, we showcase an example of the summarized meta-review critique generated for the reviews of the previously

introduced ALS mechanism hypotheses.

Research overview generation. The Meta-review agent periodically synthesizes top-ranked hypotheses into a research overview, providing a roadmap for future research. This overview outlines potential research areas and directions relevant to the research goal, justifying their importance and suggesting specific experiments within each. Each area includes illustrative example topics. The research overview also serves as an additional input to the Generation agent in subsequent iterations.

The research overview serves to effectively map the boundary of current knowledge relevant to the research goal in the co-scientist system and helps highlight future areas of exploration. In Appendix Figure A.20-A.21, we show an example of a research overview for the ALS mechanism research goal.

The Meta-review agent can further format these overviews using constrained decoding techniques [70] to adhere to common research publication and grant formats (e.g., National Institute of Health (NIH) Specific Aims Page format). We demonstrate the effectiveness of this in subsequent sections.

Research contacts identification. The Meta-review agent uses prior literature review to suggest qualified domain experts for research hypotheses and proposal review, including the reasoning behind each suggestion. These potential contacts are summarized in the research overview, providing researchers with additional perspectives and potential avenues for collaborations. An example research contact (with the researcher name redacted) is shown in Appendix Figure A.22.

3.4 Expert-in-the-loop interactions with the co-scientist

The AI co-scientist empowers scientists to actively guide the system through an expert-in-the-loop design (Figure 2). Scientists can interact with the system in several ways:

- Refine the initial research goal in light of the generated hypotheses and research overview.
- Provide manual reviews of generated hypotheses (see Section 3.3.2 for other system generated review types), which the co-scientist uses to evaluate and improve the hypotheses and proposals.
- Contribute their own hypotheses and proposals for inclusion in the tournament, where they are ranked alongside and can be combined with system-generated hypotheses and proposals.
- Direct the co-scientist to follow up on specific research directions (for example restricted to a smaller collection of prior publications). When this research is referenced in the research goal, the co-scientist can prioritize generation methods that can access and synthesize it.

3.5 Tool use in AI co-scientist

The co-scientist leverages various tools during the generation, review, and improvement of hypotheses and research proposals. Web search and retrieval are primary tools, important for grounded, up-to-date hypotheses.

For research goals that explore a constrained space of possibilities (e.g., all known cell receptors of a specific type or all FDA-approved drugs), the co-scientist agents utilize domain-specific tools, such as open databases, to constrain searches and generate hypotheses. The co-scientist can also index and search a private repository of publications specified by the scientist.

Finally, the system can utilize and incorporate feedback from specialized AI models like AlphaFold. We demonstrate this qualitatively with a protein design example in the Appendix Section A.6.